

29 — INTRODUCTION TO DEEP NEURAL NETWORKS USING KERAS

Aim

To build and train a simple deep neural network with multiple hidden layers using TensorFlow/Keras on the MNIST handwritten digit dataset.

Algorithm + Procedure

1. Import TensorFlow and Keras libraries.
2. Load the MNIST handwritten digit dataset.
3. Separate the dataset into training and testing sets.
4. Normalize the pixel values from the range 0–255 to 0–1.
5. Create a sequential neural network model.
6. Add a Flatten layer to convert the 28×28 image into a one-dimensional vector.
7. Add a Dense hidden layer with 128 neurons and ReLU activation.
8. Add another Dense hidden layer with 64 neurons and ReLU activation.
9. Add an output layer with 10 neurons and Softmax activation.
10. Compile the model using the Adam optimizer and sparse categorical cross-entropy loss.
11. Train the model for 5 epochs.
12. Evaluate the trained model using the test dataset.
13. Display the final test accuracy.

Code

```
import tensorflow as tf

from tensorflow.keras import layers, models

# Load MNIST dataset
(X_train, y_train), (X_test, y_test) = (
    tf.keras.datasets.mnist.load_data()
)

# Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# Create Deep Neural Network
```

```
model = models.Sequential([
    layers.Input(shape=(28, 28)),
    layers.Flatten(),
    layers.Dense(
        128,
        activation='relu'
    ),
    layers.Dense(
        64,
        activation='relu'
    ),
    layers.Dense(
        10,
        activation='softmax'
    )
])

# Compile model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train model
model.fit(
    X_train,
    y_train,
    epochs=5,
    validation_split=0.1
)
```

```
# Evaluate model

loss, accuracy = model.evaluate(

    X_test,

    y_test,

    verbose=0

)

print("Test accuracy:", accuracy)
```

Output

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ————— 0s 0us/step
Epoch 1/5
1688/1688 ————— 11s 5ms/step - accuracy: 0.9250 - loss: 0.2521 - val_accuracy: 0.9662 - val_loss: 0.1148
Epoch 2/5
1688/1688 ————— 9s 5ms/step - accuracy: 0.9682 - loss: 0.1065 - val_accuracy: 0.9753 - val_loss: 0.0860
Epoch 3/5
1688/1688 ————— 8s 4ms/step - accuracy: 0.9771 - loss: 0.0734 - val_accuracy: 0.9760 - val_loss: 0.0862
Epoch 4/5
1688/1688 ————— 9s 5ms/step - accuracy: 0.9824 - loss: 0.0564 - val_accuracy: 0.9752 - val_loss: 0.0885
Epoch 5/5
1688/1688 ————— 8s 5ms/step - accuracy: 0.9854 - loss: 0.0451 - val_accuracy: 0.9788 - val_loss: 0.0815
Test accuracy: 0.9763000011444092
```

Result

A deep neural network with **two hidden layers** was successfully created and trained using TensorFlow/Keras on the MNIST dataset. The model achieved approximately **97–98% test accuracy**.